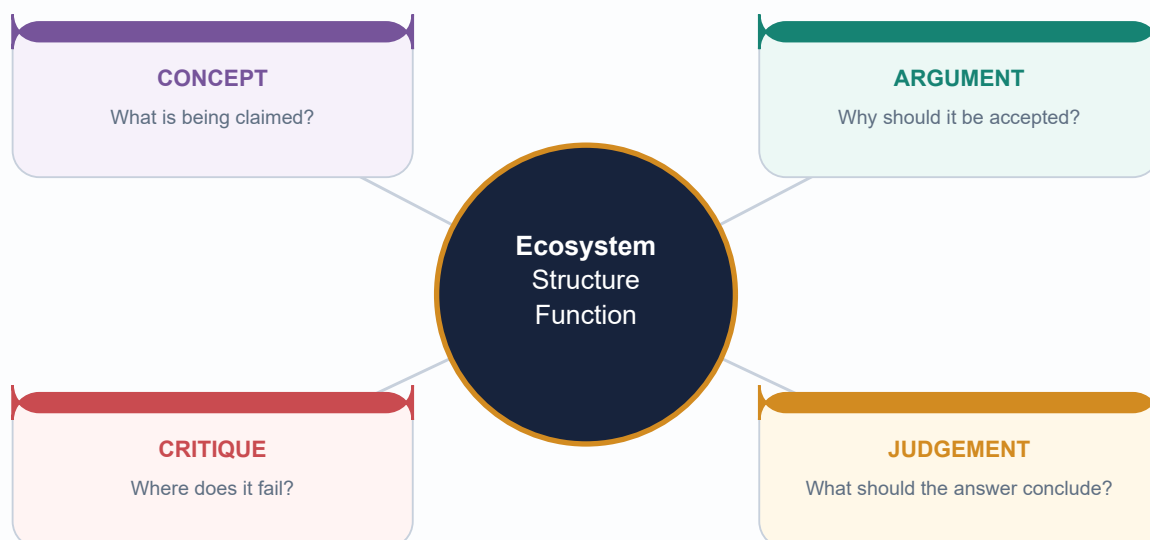


UPSC ENVIRONMENT AND ECOLOGY WORKBOOK

Ecosystem Structure and Function Solved-Practice Workbook

Subject: Environment and Ecology | Paper: GS-III
Eight source-trap MCQs with strict A -> B -> C -> D rotation
Original 10-, 15- and 20-mark solved content frameworks



Facts from official sources | Inferences clearly marked | No fabricated data

HIGH UPSC RELEVANCE

HIGH

Topic-Specific MCQ Loop 1

GS-III

Environment and
Ecology

INTRO & ORIGIN

Each question is generated from an explicit source-chapter trap and explanation.

STATIC THEORY

- Q1. Which statement correctly resolves this common misconception: "Energy cycles like nutrients in an ecosystem."?

- A. Energy flows one-way and dissipates;
- B. Energy cycles like nutrients in an ecosystem.
- C. Habitat and niche are the same thing.
- D. Decomposers are only bacteria.

- ANSWER: A - Energy flows one-way and dissipates;

- Q2. Which statement correctly resolves this common misconception: "Habitat and niche are the same thing."?

- A. Habitat and niche are the same thing.
- B. Habitat is the address; niche is the functional
- C. Decomposers are only bacteria.
- D. A food chain and a food web are interchangeable terms.

- ANSWER: B - Habitat is the address; niche is the functional

- Q3. Which statement correctly resolves this common misconception: "Decomposers are only bacteria."?

- A. Decomposers are only bacteria.
- B. A food chain and a food web are interchangeable terms.
- C. Fungi and detritivores (e.g., earthworms, dung
- D. Producers are always green plants.

- ANSWER: C - Fungi and detritivores (e.g., earthworms, dung

- Q4. Which statement correctly resolves this common misconception: "A food chain and a food web are interchangeable terms."?

- A. A food chain and a food web are interchangeable terms.
- B. Producers are always green plants.
- C. An ecotone is simply a boundary line.
- D. A food web is the sum of

- ANSWER: D - A food web is the sum of

MEMORY HOOK

Resolve the misconception before memorising the fact.

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function ->
Source-grounded practice

HIGH**Topic-Specific MCQ Loop 2**

GS-III
Environment and
Ecology

INTRO & ORIGIN

Each question is generated from an explicit source-chapter trap and explanation.

STATIC THEORY

- Q5. Which statement correctly resolves this common misconception: "Producers are always green plants."?
- A. Chemosynthetic bacteria (e.g., at hydrothermal
- B. Producers are always green plants.
- C. An ecotone is simply a boundary line.
- D. "Biomass" means the live weight of organisms.
- ANSWER: A - Chemosynthetic bacteria (e.g., at hydrothermal
- Q6. Which statement correctly resolves this common misconception: "An ecotone is simply a boundary line."?
- A. An ecotone is simply a boundary line.
- B. It is a *zone* of variable width and a zone of
- C. "Biomass" means the live weight of organisms.
- D. A stable-looking ecosystem is automatically resilient.
- ANSWER: B - It is a *zone* of variable width and a zone of
- Q7. Which statement correctly resolves this common misconception: "'Biomass' means the live weight of organisms."?
- A. "Biomass" means the live weight of organisms.
- B. A stable-looking ecosystem is automatically resilient.
- C. It is conventionally the **net dry**
- D. Energy cycles like nutrients in an ecosystem.
- ANSWER: C - It is conventionally the **net dry**
- Q8. Which statement correctly resolves this common misconception: "A stable-looking ecosystem is automatically resilient."?
- A. A stable-looking ecosystem is automatically resilient.
- B. Energy cycles like nutrients in an ecosystem.
- C. Habitat and niche are the same thing.
- D. Stability (resistance to
- ANSWER: D - Stability (resistance to

MEMORY HOOK

Resolve the misconception before memorising the fact.

MAINS ANGLE**STUDY LINK**

Ecosystem Structure and Function ->
Source-grounded practice

HIGH

Solved 10-Mark Practice

GS-III
Environment and
Ecology

INTRO & ORIGIN

Explain Ecosystem Structure and Function with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE

1. Concept

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

- ☐ **MoEFCC**: apex ministry for ecosystem/biodiversity policy and international convention
- ☐ **Wildlife Institute of India (WII), Dehradun**: research on ecosystem and species
- ☐ **Botanical Survey of India (BSI) / Zoological Survey of India (ZSI)**: taxonomic and
- ☐ ☐ Ecosystem concepts underlie every protected-area, EIA and afforestation decision, so this
- ☐ ☐ A mangrove ecotone (e.g., Sundarbans) buffers cyclones, nurseries fish and stores "blue
- ☐ ☐ A degraded grassland with a broken food web (e.g., loss of apex predators) can trigger
- ☐ ☐ Western Ghats montane forest stratification (canopy, understorey, shrub, floor) supports
- ☐ Producers are the sole entry point of energy into an ecosystem; energy flow is
- ☐ A food web is a network of interconnected food chains, giving greater ecosystem stability
- ☐ Decomposers are heterotrophic and act on dead organic matter; detritivores (e.g.,
- ☐ Ecotones typically show the "edge effect" — higher species richness than either
- ☐ Niche cannot be shared identically by two species in the same habitat (competitive
- ☐ The term "ecosystem" was coined by **A.G. Tansley (1935)**; the biosphere's abiotic
- ☐ A niche is decomposed into habitat, trophic (food) and reproductive niches — a
- ☐ **Biomass** is measured as *net dry weight* of organic matter, not fresh/wet weight —

MAINS ANGLE

These are model content points; arrange them into paragraphs rather than reproducing a list.

STUDY LINK

Ecosystem Structure and Function -> 10-mark answer

HIGH**Solved 15-Mark Practice****GS-III**Environment and
Ecology**INTRO & ORIGIN**

Analyse Ecosystem Structure and Function with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE**1. Concept**

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

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MAINS ANGLE

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STUDY LINK

Ecosystem Structure and Function -> 15-mark answer

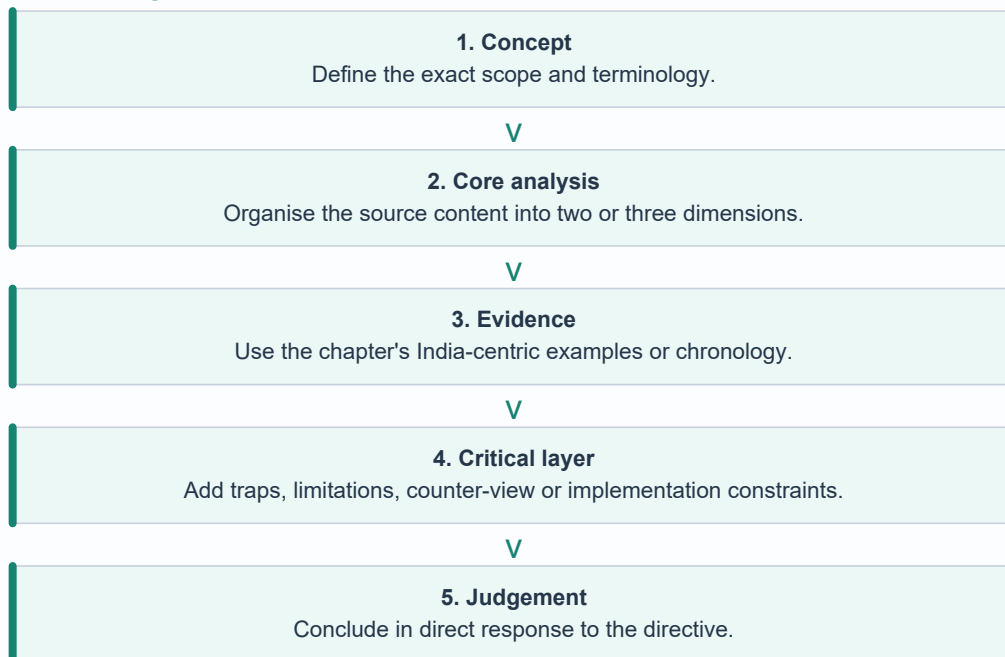
HIGH

Solved 20-Mark Practice

GS-III
Environment and
Ecology

INTRO & ORIGIN

Critically examine Ecosystem Structure and Function with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE**STATIC THEORY**

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MAINS ANGLE

These are model content points; arrange them into paragraphs rather than reproducing a list.

STUDY LINK

Ecosystem Structure and Function -> 20-mark answer

HIGH

Workbook Rapid Revision

GS-III
Environment and
Ecology

INTRO & ORIGIN

Final revision is placed after teaching and solved practice.

TOPIC RECALL MAP

Ecosystem Structure and Function

V

Definition ☐ MoEFCC: apex ministry for ecosystem/biodiversity policy and international convention	Mechanism ☐ Wildlife Institute of India (WII), Dehradun: research on ecosystem and species
Application ☐ Botanical Survey of India (BSI) / Zoological Survey of India (ZSI): taxonomic and	Evaluation ☐ ☐ Ecosystem concepts underlie every protected-area, EIA and afforestation decision, so this

STATIC THEORY

- ☐ **MoEFCC:** apex ministry for ecosystem/biodiversity policy and international convention
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MEMORY HOOK

Meaning -> mechanism -> evidence -> limitation -> judgement

MUST-KNOW FACTS

1. ☐ The term "ecosystem" was coined by **A.G. Tansley (1935)**; the biosphere's abiotic
2. ☐ A niche is decomposed into habitat, trophic (food) and reproductive niches — a
3. ☐ **Biomass** is measured as *net dry weight* of organic matter, not fresh/wet weight —
4. ☐ As of the 18th India State of Forest Report (ISFR 2023, released December 2024 — the
5. ☐ Economic Survey 2025-26 (Ch. 10, *Environment and Climate Change*) frames India's
6. ☐ ☐ Recurring Prelims pattern: identify correct statements on energy flow, trophic levels

UPSC TRAPS

- WRONG:** Energy cycles like nutrients in an ecosystem.
CORRECT: Energy flows one-way and dissipates;
- WRONG:** Habitat and niche are the same thing.
CORRECT: Habitat is the address; niche is the functional
- WRONG:** Decomposers are only bacteria.
CORRECT: Fungi and detritivores (e.g., earthworms, dung
- WRONG:** A food chain and a food web are interchangeable terms.
CORRECT: A food web is the sum of

MAINS ANGLE

Use Ecosystem Structure and Function as an argument, not as an unstructured list of facts.

STUDY LINK

Environment and Ecology -> Ecosystem
Structure and Function

End of Register Notes - UPSC Agent / Copilot CLI